Problem Summary

Given two strings s and t, determine whether t is an anagram of s.

An anagram means both strings contain the exact same frequencies, just possibly in a different order

Key Idea

Count how many times each character appears

→ add counts using s

→ subtract counts using t

If the strings are anagrams, every character count should end at 0.

Diagram Example

s = "anagram"

a: 3 n: 1 g: 1 r: 1 m: 1

t = "nangaram"

a: 0 n: 0 g: 0 r: 0 m: 0

All are 0 so anagrams

Approach

1) If the lengths are different, return False

2) Create a dictionary for counts

3) Traverse s: add 1 to each character count

4) Traverse t: subtract from each character count

5) If all values in count are 0, return True

6) Otherwise return False

Time Complexity: O(n) → each character is checked once in both strings

Space Complexity: O(k) → where k is the number of unique characters